Massachusetts Institute of Technology

MIT 16.S897 Spacecraft ADCS

Spacecraft ADCS Report

ASTERIA CubeSat Dynamics and Stability Analysis

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Changelog

Date	Section Modified	Description of Changes
Feb 10, 2026	Project Setup	Created github repository and selected Asteria mission for study.
Feb 11, 2026	Mission Overview	Defined ASTERIA mission objectives and ADCS requirements.
Feb 12, 2026	Orbital Model	Implemented two-body orbital dynamics in the ECI frame.
Feb 13, 2026	Orbital Simulation	Added RK4 integration for orbit propagation and generated trajectory plots.
Feb 14, 2026	Spacecraft Geometry	Defined body frame and decomposed spacecraft mass properties.
Feb 15, 2026	Inertia Analysis	Derived inertia matrix using the parallel axis theorem.
Feb 16, 2026	Principal Axes	Verified diagonalization and extracted principal moments.
Feb 18, 2026	Rigid-Body Dynamics	Derived torque-free Euler equations in the principal axis frame.
Feb 20, 2026	Numerical Integration	Implemented RK4 solver for rotational dynamics.
Feb 22, 2026	Stability Study	Added principal-axis perturbation tests (0.01 rad/s).
Feb 23, 2026	Time-Domain Analysis	Generated angular velocity time-history plots and interpreted stability behavior.
Feb 24, 2026	Momentum Sphere	Computed angular momentum trajectories and polhode curves.
Feb 25, 2026	Final Validation	Verified document typos, figure formatting, and sources.

References

1. Wertz, J. R., & Larson, W. J. *Space Mission Analysis and Design (SMAD)*. Microcosm Press.

2. MIT 16.S897 Lecture Notes 1–4.

3. Knapp, M., Seager, S., Demory, B. O., et al. (2020). *Demonstrating High-precision Photometry with a CubeSat: ASTERIA Observations of 55 Cancri e*. The Astronomical Journal, 160(1), 23.

4. Pong, C. M. (2018). *Pointing Control for the ASTERIA 6U CubeSat*. Proceedings of the 32nd Annual AIAA/USU Conference on Small Satellites.

5. Smith, M. W., et al. (2018). *The Arcsecond Space Telescope Enabling Research in Astrophysics (ASTERIA) Mission*. SPIE Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave.

6. Bocchino, R. L., et al. (2019). *F Prime: An Open-Source Framework for Small-Scale Flight Software Systems*. NASA Technical Reports Server.

1 Course Project Logistics

This homework initiates the semester-long spacecraft ADCS course project. A GitHub repository has been created to maintain simulation code and numerical tools.

Repository Link: `https://github.com/emilyaleman/Emizan_ADCS`

All simulations were performed using Python, and numerical integration used RK4.

2 Mission Description and ADCS Requirements

2.1 Mission Overview

This project analyzes the attitude dynamics and control requirements of **the ASTERIA (Arcsecond Space Telescope Enabling Research in Astrophysics) CubeSat mission**.

Mission Objectives

- **Demonstrate sub-arcsecond pointing stability** (< 0.5 arcseconds RMS over 20-minute observations) to enable high-precision photometry.
- **Achieve milli-Kelvin thermal stability** (± 0.01 K) at the focal plane to minimize detector noise during exoplanet transits.
- **Conduct opportunistic astrophysics** by observing known exoplanet systems, such as 55 Cancri e, to prove CubeSat-scale science capabilities.

Orbit Description

- Orbit type: **Low Earth Orbit (LEO)**
- Altitude: **400** km
- Inclination: **51.6** deg
- Orbital period: **5556** s

Orbital motion is modeled in the Earth-Centered Inertial (ECI) frame using the two-body gravitational model:

$$\ddot{r} = -\frac{\mu}{\|r\|^3} r \tag{1}$$

2.2 ADCS Overview

Attitude is defined as the relative rotation between the spacecraft body frame B and inertial frame N. Attitude lies on the Lie group:

$$SO(3)$$

Sensors

- Gyroscope (Inertial Measurement Unit)
- **Dual Star Trackers (Blue Canyon Technologies XACT)**
- **CMOS Image Sensor (functioning as a Fine Guidance Sensor)**

Actuators

- Reaction wheels (3-axis XACT internal wheels)
- Magnetorquers (for momentum desaturation)
- Thrusters (None - atmospheric drag used for eventual disposal)
- **2-Axis Piezoelectric Nanopositioning Stage (Focal Plane)**

ADCS Requirements

Requirement	Value
Pointing accuracy	± **0.00014** deg (0.5 arcsec)
Slew rate	**1.0** deg/s
Stability mode	**Two-Stage Fine Pointing Control**
Safe mode	**Sun-Pointing / Tumble Recovery**

3 Spacecraft Model

3.1 Body Frame Definition

A right-handed body-fixed coordinate frame $\mathcal{B} = \{b_1, b_2, b_3\}$ is defined based on the ASTERIA 6U CubeSat geometry:

- b_1: This is the width of the bus. The solar panels deploy from the 10×30 cm side faces, extending along this axis.
- b_2: This is the height of the bus. This axis passes through the "top" and "bottom" 20×30 cm faces.
- b_3: This is the longest dimension of the bus. The payload boresight (telescope instrument) points out of the "front" 10×20 cm face.

3.2 Aircraft Geometry

The spacecraft is decomposed into uniform-density shapes to calculate mass properties:

Component	Shape	Dimensions (m)	Mass (kg)
Bus	Rectangular prism	$0.2 \times 0.1 \times 0.3$ (b_1, b_2, b_3)	10.0
Solar panels (x2)	Flat plates	$0.3 \times 0.005 \times 0.2$ (b_1, b_2, b_3)	1.0 (0.5 each)

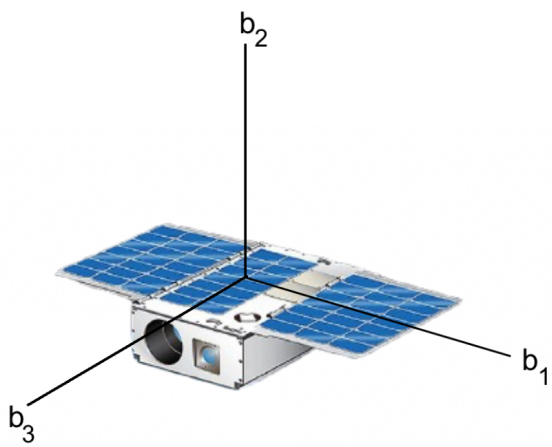

Figure 1: Asteria Body Frame Definition

The total inertia J_{total} is calculated by summing the local inertia of each component J_i translated to the system center of mass via the parallel axis theorem:

$$J_{total} = \sum_i \left(J_i + m_i (r_i^T r_i \mathbb{I} - r_i r_i^T) \right) \tag{2}$$

To understand the notation of J_{total}, we asked an LLM to decompose the transformation. The vector $r_i = [x, y, z]^T$ represents the position of the component relative to the system center of mass.

- The Dot Product $r^T r$: This result is a scalar representing the squared distance from the origin:

  $$r^T r = x^2 + y^2 + z^2$$

  Multiplying this by the identity matrix $\mathbb{I}$ distributes this total distance across the diagonal:

  $$r^T r \mathbb{I} = \begin{bmatrix} x^2 + y^2 + z^2 & 0 & 0 \\ 0 & x^2 + y^2 + z^2 & 0 \\ 0 & 0 & x^2 + y^2 + z^2 \end{bmatrix}$$

- The Outer Product (rr^T): This operation projects the displacement onto the coordinate axes, creating a symmetric matrix:

  $$rr^T = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix} = \begin{bmatrix} x^2 & xy & xz \\ yx & y^2 & yz \\ zx & zy & zz \end{bmatrix}$$

- The Resulting Translation Matrix (ΔJ): Subtracting the outer product from the identity term yields the final translation matrix:

  $$\Delta J = m \begin{bmatrix} (y^2 + z^2) & -xy & -xz \\ -yx & (x^2 + z^2) & -yz \\ -zx & -zy & (x^2 + y^2) \end{bmatrix}$$

3.3 Center of Mass

Based on the symmetrical deployment of the solar panels along the b_1 axis, the center of mass remains at the geometric center of the bus:

$$r_{cm} = \frac{1}{M} \sum m_i r_i = [0.0, 0.0, 0.0]^T \text{ m} \tag{3}$$

3.4 Inertia Matrix Calculation

The calculation is performed in three steps:

Step 1: Local Bus Inertia (J_{bus})
For a rectangular prism with mass $m = 10$ and dimensions $d = [0.2, 0.1, 0.3]$:

$$J_{11} = \frac{1}{12} M(d_2^2 + d_3^2) = \frac{10}{12}(0.1^2 + 0.3^2) = 0.0833$$

$$J_{22} = \frac{1}{12} M(d_1^2 + d_3^2) = \frac{10}{12}(0.2^2 + 0.3^2) = 0.1083$$

$$J_{33} = \frac{1}{12} M(d_1^2 + d_2^2) = \frac{10}{12}(0.2^2 + 0.1^2) = 0.0417$$

Step 2: Local Panel Inertia (J_{panel})
For one panel with mass $m = 0.5$ and dimensions $d = [0.3, 0.005, 0.2]$:

$$J_{p11} = \frac{0.5}{12}(0.005^2 + 0.2^2) \approx 0.0017$$

$$J_{p22} = \frac{0.5}{12}(0.3^2 + 0.2^2) \approx 0.0054$$

$$J_{p33} = \frac{0.5}{12}(0.3^2 + 0.005^2) \approx 0.0038$$

Step 3: Parallel Axis Translation

The solar panels are moved out along the x-axis by 0.25 m ($r = [x, 0, 0]^T$). We use the parallel axis theorem to calculate how this "offset" increases the total inertia.

Conceptually, the matrix subtraction $m(r^T r \mathbb{I} - r r^T)$ works like this:

1. We started with the squared distance (x^2) in every diagonal. This represents the maximum potential inertia added.

2. We then subtracted the axis of displacement b_1. Since the panel is moved along the x-axis, it doesn't get any harder to rotate the satellite around the x-axis. Thus, the inertia J_{11} remains unchanged. However, the displacement does affect the moments of inertia about the b_2 and b_3 axes.

$$\Delta J = 0.5 \left(\underbrace{\begin{bmatrix} x^2 & 0 & 0 \\ 0 & x^2 & 0 \\ 0 & 0 & x^2 \end{bmatrix}}_{\text{Total Distance}} - \underbrace{\begin{bmatrix} x^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{The X-Axis}} \right) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & mx^2 & 0 \\ 0 & 0 & mx^2 \end{bmatrix}$$

With $m = 0.5$ kg and $x = 0.25$ m, the added inertia for each panel is:

$$0.5(0.25^2) = 0.03125 \text{ kg} \cdot \text{m}^2 \text{ for the } J_{22} \text{ and } J_{33} \text{ terms.}$$

Final Summation:

$$J_{total} = J_{bus} + 2(J_{panel,local}) + 2(\Delta J)$$

$$J = \begin{bmatrix} 0.0867 & 0.0 & 0.0 \\ 0.0 & 0.1817 & 0.0 \\ 0.0 & 0.0 & 0.1117 \end{bmatrix}$$

3.5 Principal Axes

Because J is diagonal, the body frame axes are the principal axes ($V = \mathbb{I}$). The principal moments are:

$$\Lambda = \text{diag}(0.0867, 0.1817, 0.1117)$$

4 Orbital Dynamics

The ASTERIA mission operates in a Low Earth Orbit (LEO), having been deployed from the International Space Station. The motion is modeled in the Earth-Centered Inertial (ECI) frame.

State Vector and Governing Equations

The state vector $x(t) \in \mathbb{R}^6$ represents the position and velocity of the CubeSat:

$$x = \begin{bmatrix} r \\ v \end{bmatrix} = \begin{bmatrix} r_x & r_y & r_z & v_x & v_y & v_z \end{bmatrix}^T$$

The dynamics are governed by the following first-order ODE, assuming a point-mass Earth (Two-Body Problem):

$$\dot{x} = \begin{bmatrix} \dot{r} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ -\frac{\mu}{\|r\|^3} r \end{bmatrix} \tag{4}$$

Where:

- $\mu = 398600.4418$ km^3/s^2 (Earth's gravitational parameter).
- r is the position vector from the center of the Earth.
- v is the inertial velocity vector.

Simulation Initial Conditions

To simulate ASTERIA's orbit (based on an ISS-like deployment at ~ 400 km altitude), we use:

- $R_e = 6378.137$ km (Earth Equatorial Radius)
- $r_0 = [R_e + 400, 0, 0]^T$ km
- $v_0 = [0, \sqrt{\mu/\|r_0\|}, 0]^T$ km/s $\approx [0, 7.67, 0]^T$ km/s

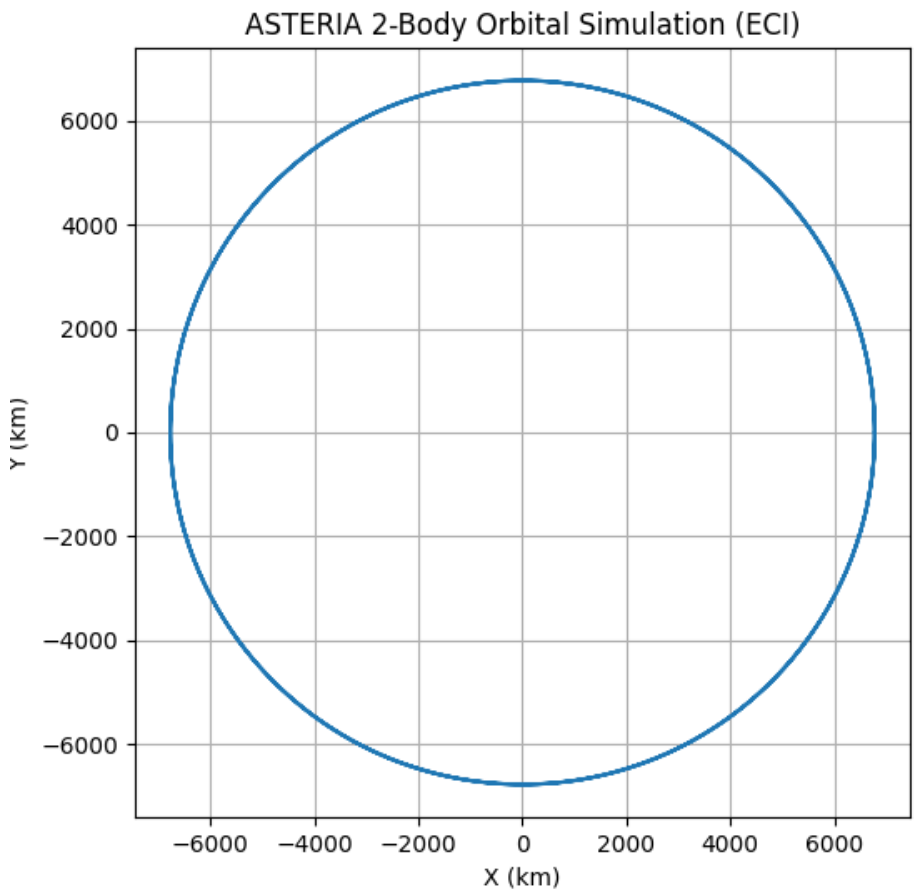

Figure 2: Numerical simulation of the ASTERIA spacecraft's orbital trajectory over three orbital periods in the Earth-Centered Inertial (ECI) frame.

5 Rigid-Body Dynamics and Euler's Equation

The rotational motion of the spacecraft is governed by Euler's rigid body equation,

$$\tau = J\dot{\omega} + \omega \times (J\omega), \tag{5}$$

where J is the inertia tensor and ω is the angular velocity vector expressed in the body frame.

For this analysis, we consider the torque-free case ($\tau = 0$), because it allowed us to study the inherent stability of the satellite's geometry without the influence of external forces. This gives us

$$J\dot{\omega} + \omega \times (J\omega) = 0. \tag{6}$$

The inertia tensor J is a second-order tensor relating angular velocity to angular momentum,

$$\mathbf{H} = J\boldsymbol{\omega}. \tag{7}$$

In a general body frame, J may contain off-diagonal terms,

$$J = \begin{bmatrix} J_{11} & J_{12} & J_{13} \\ J_{21} & J_{22} & J_{23} \\ J_{31} & J_{32} & J_{33} \end{bmatrix}. \tag{8}$$

However, by selecting the principal axis frame, the inertia tensor becomes diagonal,

$$J = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}. \tag{9}$$

Let

$$\boldsymbol{\omega} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}. \tag{10}$$

Substituting the diagonal inertia tensor into the torque-free equation and evaluating the cross product yields the scalar Euler equations:

$$I_1\dot{\omega}_1 = (I_2 - I_3)\omega_2\omega_3, \tag{11}$$
$$I_2\dot{\omega}_2 = (I_3 - I_1)\omega_3\omega_1, \tag{12}$$
$$I_3\dot{\omega}_3 = (I_1 - I_2)\omega_1\omega_2. \tag{13}$$

These coupled nonlinear differential equations describe the time evolution of the spacecraft's angular velocity components under torque-free motion.

5.1 Stability of Principal Axis Rotations

The system of nonlinear ordinary differential equations was integrated using a fourth-order Runge–Kutta (RK4) method with fixed time step Δt. Simulations were initialized with a total angular velocity magnitude of:

$$|\omega| = 10 \text{ RPM}$$

To test the robustness of each axis, a small perturbation of 0.01 rad/s was added to the transverse components. The results verify the Intermediate Axis Theorem (Dzhanibekov Effect) for the spacecraft's specific inertia distribution:

- **Minor axis ($J_1 = 0.0867$): Marginally stable.**
- **Major axis ($J_2 = 0.1817$): Marginally stable.**
- **Intermediate axis ($J_3 = 0.1117$): Unstable.**

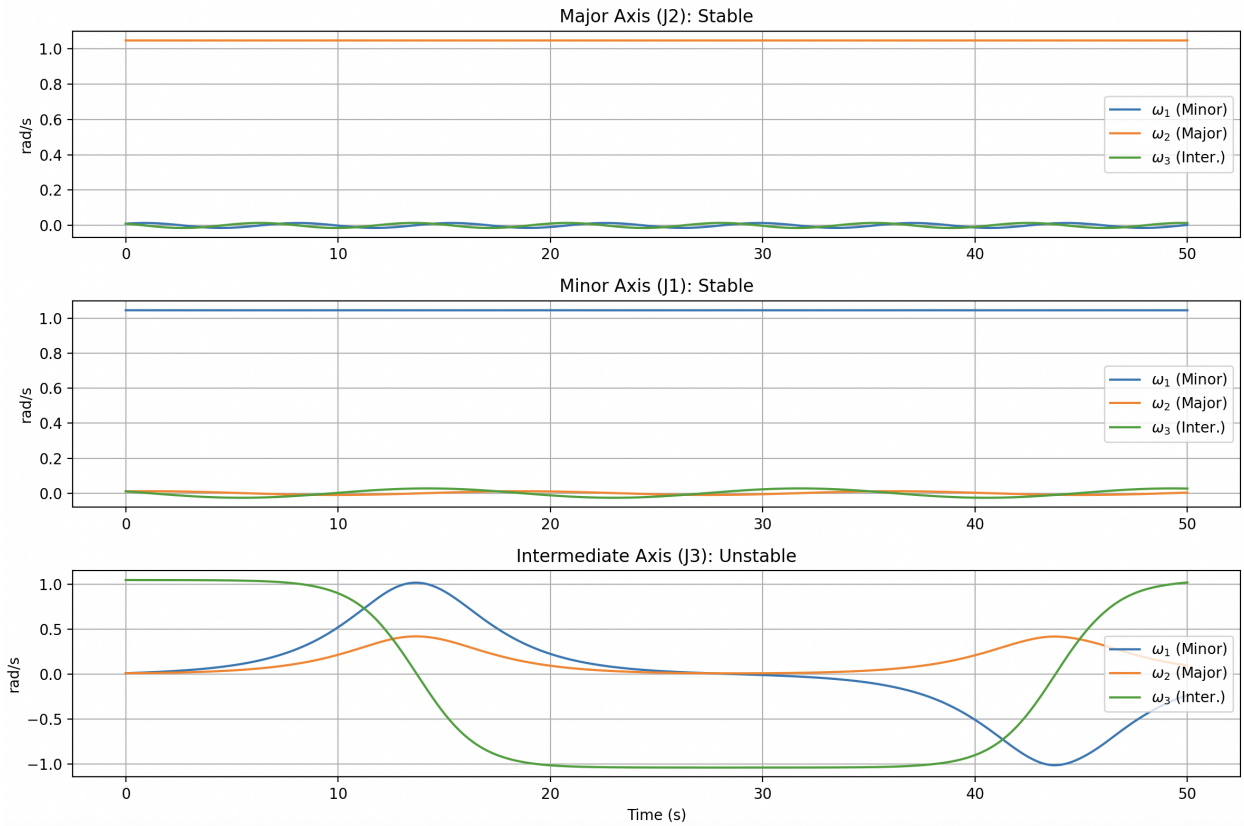

Figure 3: Angular velocity components versus time for torque-free rotation initialized at 10 RPM with a .01 perturbation. The intermediate principal axis exhibits instability, while the minimum and maximum axes remain stable.

5.2 Momentum Sphere Visualization

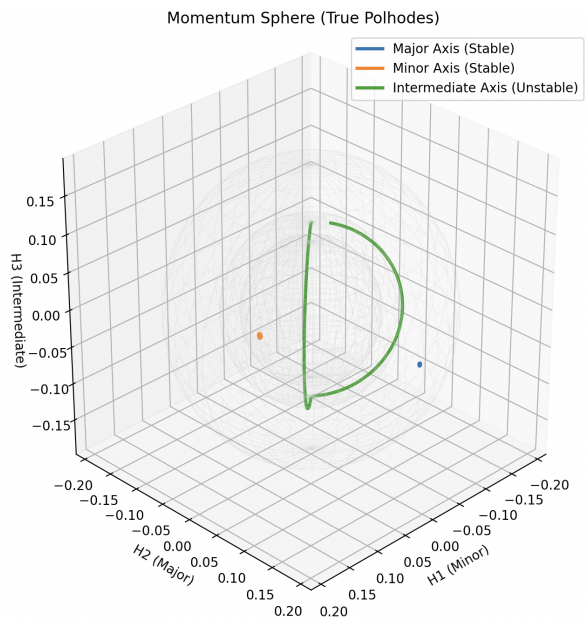

Figure 4: Momentum Sphere and Polhode Trajectories for the Spacecraft Inertia Tensor, analogous to a gyroscope

As shown in Figure 4, rotations about the minor and major principal axes produce closed polhode trajectories on the momentum sphere. These closed curves indicate that small perturbations remain bounded over time, consistent with stable torque-free motion. In contrast, the intermediate principal axis does not produce a closed trajectory under perturbation. Instead, small deviations grow over time, causing the angular velocity vector to move away from the initial configuration. This behavior confirms the inherent instability predicted by the Intermediate Axis Theorem.

This geometric interpretation is similar to the behavior of a torque-free gyroscope and helped us visualize the spacecraft's rotational stability properties.

6 Conclusions

In this project, we developed and implemented the foundational orbital and rotational dynamics models for the ASTERIA CubeSat and used them to better understand how the spacecraft behaves in space. By modeling the 6U structure and deployed solar arrays, we derived the full inertia matrix and computed the principal moments of inertia as $\Lambda = \mathrm{diag}(0.0867, 0.1817, 0.1117)\ \mathrm{kg} \cdot \mathrm{m}^2$, confirming that the body frame is closely aligned with the principal axes. Using a two-body gravitational model with an adaptive RK45 solver, we numerically propagated the orbit and verified stable motion at an altitude of approximately 400 km, consistent with deployment from the ISS. We then implemented Euler's torque-free rotational equations with a custom RK4 integrator to analyze the spacecraft's attitude behavior. Our simulations showed that rotations about the minor axis (b_1, $0.0867\ \mathrm{kg} \cdot \mathrm{m}^2$) and major axis ($b_2$, $0.1817\ \mathrm{kg} \cdot \mathrm{m}^2$) remained bounded and stable, while rotation about the intermediate axis (b_3, $0.1117\ \mathrm{kg} \cdot \mathrm{m}^2$) became unstable under a small perturbation of only 0.01 rad/s, consistent with the Intermediate Axis Theorem. This result is especially significant because ASTERIA's telescope boresight aligns with the unstable axis, meaning passive stability

alone is insufficient. Overall, this project helped us connect rigid body dynamics theory to a real spacecraft system, reinforced concepts like principal moments, torque-free motion, and orbital propagation, and clearly demonstrated why a robust active ADCS is required to maintain precise pointing performance in Low Earth Orbit.